Variation of parameters:

Let $f(D)y = y^{(n)} + P_{n-1}(x)y^{(n-1)} + \cdots + P_1(x)y'(x) + P_0(x)y = R(x)$,
where $P_{n-1}, \ldots, P_1, P_0, R$ are continuous on open interval I.

Suppose $y_1, y_2, \ldots, y_n$ are n- L.I. solutions of homogeneous part $f(D)y = 0$ on I.

$\therefore$ C.F $= c_1y_1 + c_2y_2 + \cdots + c_ny_n$, c_i's are constants.

Then $\boxed{y_p = c_1(x)y_1 + c_2(x)y_2 + \cdots + c_n(x)y_n}$ on I, where $c_1(x), c_2(x), \ldots c_n(x)$ are functions of x given by

$$\boxed{c_1(x) = \int \frac{W_1(x)}{W(x)} R(x)\,dx, \quad \ldots\ldots \quad c_n(x) = \int \frac{W_n(x)}{W(x)} R(x)\,dx,}$$

$W(x)$ is the Wronskian of $y_1, y_2, \ldots, y_n$ and W_i is obtained from $W(x)$ by replacing ith column of W by the column $\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$.

Ex. Derive the above formula for $n=2$.

Sketch of Proof: $y'' + P_1(x)y' + P_2(x)y = R(x)$.

Suppose C.F $= c_1y_1 + c_2y_2$.

Take $y_p = c_1(x)y_1 + c_2(x)y_2$.

$y_p' = c_1(x)y_1' + c_1'(x)y_1 + c_2y_2' + c_2'y_2$.

Set $c_1'(x)y_1 + c_2'(x)y_2 = 0$. —— (1)

$y_p' = c_1(x)y_1' + c_2(x)y_2'$, $y_p'' = ?$

Find y_p'' and put in the given eq., this will give

$c_1'(x)y_1' + c_2'(x)y_2' = R(x)$ —— (2). *

Write the eq.s (1) & (2) in the matrix form

$$\begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix} \begin{bmatrix} c_1'(x) \\ c_2'(x) \end{bmatrix} = \begin{bmatrix} 0 \\ R(x) \end{bmatrix}.$$

$$\Rightarrow \quad c_1(x) = \int \frac{W_1}{W} R(x)\,dx, \qquad c_2(x) = \int \frac{W_2}{W} R(x)\,dx,$$

$$W_1 = \det\begin{bmatrix} 0 & y_2 \\ 1 & y_2' \end{bmatrix} = -y_2, \qquad W_2 = \det\begin{bmatrix} y_1 & 0 \\ y_1' & 1 \end{bmatrix} = y_1.$$

$$\therefore \boxed{y_p = -y_1 \int \frac{y_2}{W} R(x)\,dx + y_2 \int \frac{y_1}{W} R(x)\,dx.}$$

Ex.1. $y'' + 9y = \sec 3x.$

Sol: $C.F = c_1 \cos 3x + c_2 \sin 3x$, $\quad y_1 = \cos 3x,\ y_2 = \sin 3x.$

By the formula of variation of parameter,

$$y_p = -\cos 3x \int \frac{\sin 3x}{W} \sec 3x\,dx + \sin 3x \int \frac{\cos 3x}{W} \sec 3x\,dx,$$

$$W = \det\begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix} = \det\begin{bmatrix} \cos 3x & \sin 3x \\ -3\sin 3x & 3\cos 3x \end{bmatrix}$$

$$= 3.$$

$$\therefore \quad y_p = -\cos 3x \int \frac{\tan 3x}{3}\,dx + \sin 3x \int \frac{1}{3}\,dx$$

$$= +\frac{1}{3}\cos 3x \frac{\ln \cos 3x}{3} + \frac{\sin 3x}{3} \times x$$

$$\boxed{y_p = \frac{1}{9}\cos 3x \ln \cos 3x + \frac{1}{3} x \sin 3x.}$$

General sol $y = C.F. + P.I.$

Ex: By the method of variation of parameters, find a particular integral of the following diff eqs.

(i) $y'' + y = \operatorname{cosec} x$ $\qquad$ (ii) $y'' + 4y = 3\cos 2x$

(iii) $y'' + 3y' + 2y = \dfrac{1}{1+e^x}$ $\qquad$ (iv) $y'' - y' - 2y = e^{-x}$

(v) $(x^2-1)y'' + 4xy' + 2y = \frac{2}{(x+1)}$, $y(0) = -1$, $y'(0) = -5$,

given that $y_1 = \frac{1}{x+1}$ is a solution of homogeneous part.

$\longrightarrow$

Ex. $y''' - 2y'' - 4y' + 8y = e^{-3x} + 8x^2.$

Sol^n: Put $y = e^{mx}$ in $y''' - 2y'' - 4y' + 8y = 0$,

$$m^3 - 2m^2 - 4m + 8 = 0$$
$$\Rightarrow m^2(m-2) - 4(m-2) = 0$$
$$\Rightarrow (m^2-4)(m-2) = 0$$
$$\Rightarrow m = 2, 2, -2$$

$\therefore$ C.F. $= c_1 e^{2x} + c_2 x e^{2x} + c_3 e^{-2x}$.

Take $y_1 = e^{2x}$, $y_2 = xe^{2x}$, $y_3 = e^{-2x}$

By the method of variation of parameters,

$$y_p = c_1(x)\, e^{2x} + c_2(x)\, xe^{2x} + c_3(x)\, e^{-2x},$$

$$c_1(x) = \int \frac{W_1}{W}\left(e^{-3x} + 8x^2\right) dx, \quad c_2(x) = \int \frac{W_2}{W}\left(e^{-3x} + 8x^2\right) dx$$

$$c_3(x) = \int \frac{W_3}{W}\left(e^{-2x} + 8x^2\right) dx.$$

$$W = \det\begin{bmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{bmatrix} = \det\begin{bmatrix} e^{2x} & xe^{2x} & e^{-2x} \\ 2e^{2x} & (2x+1)e^{2x} & -2e^{-2x} \\ 4e^{2x} & (4x+4)e^{2x} & 4e^{-2x} \end{bmatrix}$$

$$= e^{2x} \det\begin{bmatrix} 1 & x & 1 \\ 2 & (2x+1) & -2 \\ 4 & 4(x+1) & 4 \end{bmatrix}$$

$$= e^{2x}\left\{ 8x+4 + 8x + 8 - x\left[8+8\right] + 8x + 8 - 8x - 4 \right\}$$

$$= 16e^{2x}.$$

$$W_1 = \det\begin{bmatrix} 0 & xe^{2x} & e^{-2x} \\ 0 & (2x+1)e^{2x} & -2e^{-2x} \\ 1 & 4(x+1)e^{2x} & 4e^{-2x} \end{bmatrix} = -2x - 2x - 1 = -4x - 1$$

$$W_2 = \det\begin{bmatrix} e^{2x} & 0 & e^{-2x} \\ 2e^{2x} & 0 & -2e^{-2x} \\ 4e^{2x} & 1 & 4e^{-2x} \end{bmatrix} = -(-2-2) = 4$$

$$W_3 = \det \begin{bmatrix} e^{2x} & xe^{2x} & 0 \\ 2e^{2x} & (2x+1)e^{2x} & 0 \\ 4e^{2x} & 4(x+1)e^{2x} & 1 \end{bmatrix} = e^{4x}(2x+1-2x)$$

$$= e^{4x}.$$

$$\therefore \quad C_1(x) = \int -\frac{(4x+1)}{16e^{2x}} \left(e^{-3x} + 8x^2\right) dx$$

$$= \int -\frac{(4x+1)}{16} e^{-5x} - \frac{(4x+1)}{2} x^2 e^{-2x} dx$$

$$= -\frac{1}{16}\left[(4x+1)\frac{e^{-5x}}{-5} - \int 4\frac{e^{-5x}}{-5}\right] dx$$

$$- \frac{1}{2}\left[\frac{(4x^3+x^2)e^{-2x}}{-2} - \int \frac{(12x^2+2x)e^{-2x}}{-2} dx\right]$$

$$= \frac{1}{80}\left[(4x+1)e^{-5x} + \frac{4}{5}e^{-5x}\right]$$

$$+ \frac{1}{4}\left[(4x^3+x^2)e^{-2x} - \int (12x^2+2x)e^{-2x} dx\right]$$

(Ex.)

$$C_2(x) = \int \frac{4}{16e^{2x}} \left(e^{-3x} + 8x^2\right) dx$$

$$= \frac{1}{4}\int e^{-5x} + 8x^2 e^{-2x} dx. \qquad \text{(Ex.)}$$

$$C_3(x) = \int \frac{e^{4x}}{16e^{2x}} \left(e^{-3x} + 8x^2\right) dx$$

$$= \frac{1}{16}\int e^{-x} + 8x^2 e^{2x} dx \quad - \text{(Ex.)}$$

$$y_p = C_1(x)e^{2x} + C_2(x)\,xe^{2x} + C_3(x)e^{-2x}.$$

General sol^n $y = C.F. + P.I.$